DANIEL BEECHEY

Email ◇ Google Scholar ◇ Website ◇ GitHub

INTERESTS

Reinforcement Learning, LLM-based Agents, Explainable AI, Continual Learning, Self-Evolution, Exploration, Open-ended Learning, Bounded Rationality

EDUCATION

PhD in Reinforcement Learning *2026*
University of Bath, UK
Supervisors: Özgür Şimşek (Computer Science), Emma Carmel (Social Policy)
Dissertation: *Explaining Reinforcement Learning with Shapley Values: Theory and Algorithms*

MRes in Accountable, Responsible, and Transparent AI *2022*
University of Bath, UK
Dissertation: *Explaining Reinforcement Learning with Shapley Values*
Grade: Distinction

MSc in Data Science *2021*
University of Bath, UK
Dissertation: *Autonomous Routing of Printed Circuit Boards with Hierarchical Reinforcement Learning*
Grade: Distinction

BSc (Hons) in Mathematics *2020*
University of Bath, UK
Grade: First Class

EXPERIENCE

Core Researcher, *H Company, London* *2026 - present*
Using RL to train computer-use agents.

Research Intern, *H Company, London* *2026 - 2026*
Using RL to train computer-use agents.

Research Scientist, *Huawei Noah's Ark Lab, London* *2025 - 2026*
Led a research team focused on open-ended reinforcement learning for LLM-based agents.
Developed *Darwin Mobile Agent*, the first fully open-source, end-to-end pipeline for large-scale online reinforcement learning and inference of mobile GUI agents.

Co-Manager of the Bath RL Lab, *University of Bath* *2023 - 2025*
Organised lab activities, including weekly lab meetings, research sessions, paper discussions, and social events.

Teaching Assistant, *University of Bath* *2020 - 2025*
Modules: Reinforcement Learning; Statistics for Data Science; Software Technologies for Data Science; Programming, Foundations, and Connections; Programming and Discrete Mathematics; Mathematical Methods and Applications
Supervised 10 MSc and 5 BSc students.

AI Lecturer, *University of Bath* *2022 - 2023*
Lectured MSc Reinforcement Learning.
Supervised 5 MSc and 2 BSc students.

AWARDS

University of Bath, **Doctoral Recognition Award** *2024*
Bath Conference of Computer Science, **Best Overall Contribution** *2023*
Inter-CDT Conference on AI, **Best Poster** *2023*

PUBLICATIONS

Darwin Mobile Agent: A Roadmap for Self-Evolution *Preprint, 2025*
Daniel Beechey, Derek Yuen, Jianheng Liu, et al.

A Theoretical Framework for Explaining Reinforcement Learning with Shapley Values *Preprint, 2025*
Daniel Beechey, Thomas M. S. Smith, and Özgür Şimşek

Approximating Shapley Explanations in Reinforcement Learning *NeurIPS, 2025*
Daniel Beechey and Özgür Şimşek

Reformulating Reactivity Design for Data-Efficient Machine Learning *ACS Catalysis, 2023*
Toby Lewis-Atwell, **Daniel Beechey**, et al.

Explaining Reinforcement Learning with Shapley Values *ICML, 2023*
Daniel Beechey, Thomas M. S. Smith, and Özgür Şimşek

OPEN-SOURCE PROJECTS

`Darwin Mobile Agent`: An end-to-end pipeline for large-scale online reinforcement learning.

`FastSVERL`: A scalable library for approximating Shapley value explanations in reinforcement learning.

TALKS

Explaining Reinforcement Learning with Shapley Values: Theory and Algorithms
Mila RL Sofa *Sep 2026*
Cohere Labs Open Science Community Talks *Aug 2026*
MARBLE Research Group, University of Edinburgh *Sep 2025*

A Theoretical Framework for Explaining Reinforcement Learning with Shapley Values
ART-AI Colloquium Series *Feb 2025*
Bath Doctoral Festival of Ideas *Jul 2024*

An Introduction to Explainable and Hierarchical Reinforcement Learning
Bath AI Society *Apr 2024*

Explaining Reinforcement Learning with Shapley Values
Bath Conference of Computer Science *Jul 2023*
Alan Turing Institute *Jun 2023*

TECHNICAL SKILLS

Frameworks & Libraries	PyTorch, HF Transformers, VeRL, Ray, Gym, NumPy, Matplotlib
Programming Languages	Python, Bash, R, Matlab
Tools & Platforms	Git, Linux, SLURM, wandb, Jupyter, Conda
RL Algorithms	PPO, GRPO, DQN, DDPG, SAC, Hierarchical RL
Model Architectures	LLMs, Transformers, VAE, CNN, MLP

REVIEWING

Reinforcement Learning Conference (RLC) *2025, 2026*
European Workshop on Reinforcement Learning (EWRL) *2024*